

7	(a)	(i)	LEDs, which is a type of diode, will only <u>produce light when it is forward biased</u> with the A.C. power supply.	B1
			Marker's comment Many answers did not address the reverse-biasedness of the LED when an a.c. voltage is applied across the LED. This is the key to why the LED cannot operate at full power. There were many answers with correct Physics but inevitably did not directly address the question e.g. "the r.m.s. value of the a.c. is less than the maximum value", "the a.c. causes the direction of the current to change", etc. No credit was given to such responses.	
		(ii)	A <u>milliammeter</u> can be connected in series to the e.m.f. / LED to determine the threshold potential difference when the current first become non-zero when the resistance of the variable resistor is varied.	B1
			Marker's comment A vast majority of answers suggested the use of a voltmeter in the circuit. No credit is given here. The question is not asking how to measure p.d. Rather it is how to determine the threshold p.d. as a visual observation of light emitted by LED may be inaccurate/difficult. Thus we require a current-measuring device in the circuit to determine whether current flows through the LED.	
		(iii)	$V = \frac{R}{R_{total}} \times E$ $1.66 = \frac{R}{500 + R} \times 9.0$ $R = 113 \approx 110 \Omega$	C1 A1
			Marker's comment Generally well done although there were some students who wrongly thought the entire 9 V p.d. appears across the 500 Ω resistor	
	(b)		Energy released by electron = Energy of photon Since $V = \frac{\text{Work done}}{\text{charge}}$, Energy released by electron = $qV = eV$ $eV = \frac{hc}{\lambda}$ $V = \frac{hc}{e} \times \frac{1}{\lambda}$	B1 B1 A0
			Marker's comment This was very poorly done. Many students did not provide the definition of p.d. For those that did, they did not explain how work done/energy change is related to the energy of the photon. Students should take cue from the question stem which states that "During the recombination process, the energy that the electron had previously gained is released in the form of a photon" to then state that the energy change is from electrical energy (electron's) to photon energy.	

(c)	(i)	$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$ $= \frac{2.40 - 1.42}{2.31 \times 10^6 - 1.60 \times 10^6}$ $= 1.38 \times 10^{-6} \text{ V m}$	C1 A1
		Marker's comment Many students correctly avoided the use of data points in determining the gradient. However, many ignored the 10^6 order of magnitude for the $\frac{1}{\lambda}$-axis. Furthermore, many answers did not contain units.	
	(ii)	$\text{gradient} = \frac{hc}{e}$ $1.38 \times 10^{-6} = \frac{h \times 3.0 \times 10^8}{1.60 \times 10^{-19}}$ $h = 7.47 \times 10^{-34} \text{ J s}$	A1
		Marker's comment A significant number of students did not use the gradient from the earlier part to answer this part.	
(d)		There are some (small) energy losses in the LEDs due to its resistance.	B1
		Marker's comment This was poorly done. The answer is derived from the possibility that the electron did not lose all its energy to form a photon. Some of the electron energy could have gone to making the LED hotter.	

9	(a)	Work = Increase in the GPE in raising the mass to $1/4$ of the height	
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